



# 7232 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

Cycle: 4, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                     | <i>Institution</i>                                |
|---------------------------------|---------------------------------------------------|
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## OBSERVATIONS

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i> | <i>Observing Template</i> | <i>Science Target</i> |
|--------------------|--------------------|--------------|---------------------------|-----------------------|
| Observation Folder |                    |              |                           |                       |
|                    | 1                  | DDO190       | NIRCam Imaging            | (1) UGC-9240          |
|                    | 2                  | IC4662       | NIRCam Imaging            | (2) IC-4662           |
|                    | 3                  | DDO187       | NIRCam Imaging            | (3) UGC-9128          |
|                    | 4                  | GR8          | NIRCam Imaging            | (4) UGC-8091          |
|                    | 5                  | KKH86        | NIRCam Imaging            | (5) LEDA-2807150      |
|                    | 6                  | IC5152       | NIRCam Imaging            | (6) IC-5152           |
|                    | 7                  | NGC6789      | NIRCam Imaging            | (7) NGC-6789          |

## ABSTRACT

Dwarf galaxies, because of their low masses and shallow potential wells, are ideal laboratories to study the processes driving galaxy formation and evolution. However, a detailed picture of dwarf galaxy evolution is currently limited to the Local Group, making it impossible to isolate environmentally-driven effects, which play a role far beyond the host galaxy virial radius ( $D > 1$  Mpc) based on both observations and simulations.

We propose to isolate the role of environment using a carefully selected control sample of seven isolated "orphan" dwarf galaxies at  $D > \sim 2$  Mpc from the Local Group barycenter. By providing the first set of homogeneous lifetime star formation histories (SFHs) for isolated dwarfs, we will facilitate a direct, empirical comparison to their Local Group counterparts to address the following questions for the first time:

- 1) Are the SFHs of isolated dwarfs similar to each other, and/or to their less isolated counterparts? We will test the hypothesis that the diversity of dwarf SFHs near the edge of the Local Group is a result of diverse birth conditions, confirming or refuting predictions that isolated dwarfs assemble their mass both later and more uniformly than those in groups.
- 2) How far does the influence of a massive host extend? We will quantify predicted correlations between mass assembly timescales and isolation out to  $>3$  Mpc.
- 3) Do isolated dwarfs show "gappy" SFHs, predicted by simulations due to cosmic web stripping by the intergalactic medium? Post-reionization SFH "pauses" are observed for some dwarfs near the edge of the Local Group, but an environmental origin for these features can only be ruled out with a more isolated sample.

## **OBSERVING DESCRIPTION**

We use F090W+F150W for NIRCam SW imaging, providing the best compromise between color baseline and throughput for SFH fitting. S/N requirements are set by the need to reach  $S/N=10$  at 2 mag below the red clump of each target. S/N calculations are based on the per-filter aperture sizes recommended by Savino et al. (2024) based on JWST ERS imaging and photometry using the same dode that we employ (DOLPHOT). Given a range of target distances, our observation strategy changes slightly on a per-target basis. However, in all cases, we require a minimum of 5 groups/integration to mitigate cosmic rays, and a minimum of 4 dithers to mitigate snowball artifacts, changing integrations/exposure (and/or increasing groups/integration or number of dithers) on a per-target basis.

Targets are placed on NIRCam module B (in the corner of chip B4) as recommended to avoid larger areas of bad pixels on Module A close to the target galaxy centers. Where feasible, we use either ORIENT constraints and/or a POS TARG offset to align the NIRCam field of view with the galaxy major axis while avoiding bright foreground stars. For two of the three targets with background variability  $>5\%$ , we request a background limited constraint of  $<10\%$ , while for the third (GR8), a background constraint was prohibitive to schedulability, so to allow a POS TARG constraint we imposed no background requirement, noting that the predicted background variability was 6.1%.

# Proposal 7232 - Targets - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

| Fixed Targets | #   | Name         | Target Coordinates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Targ. Coord. Corrections                                                                                                          | Miscellaneous |
|---------------|-----|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|
|               | (1) | UGC-9240     | RA: 14 24 43.2100 (216.1800417d)<br>Dec: +44 31 41.80 (44.52828d)<br>Equinox: J2000<br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=Galaxy</i><br><i>Description=[Dwarf irregular galaxies]</i>                                                                                                                                                                                                                                                                                                          | Epoch of Position: 2000                                                                                                           |               |
|               | (2) | IC-4662      | RA: 17 47 9.1920 (266.7883000d)<br>Dec: -64 38 47.90 (-64.64664d)<br>Equinox: J2000<br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=Galaxy</i><br><i>Description=[Compact dwarf galaxy]</i>                                                                                                                                                                                                                                                                                                              | Epoch of Position: 2000                                                                                                           |               |
|               | (3) | UGC-9128     | RA: 14 15 57.0834 (213.9878475d)<br>Dec: +23 03 29.87 (23.05830d)<br>Equinox: J2000<br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=Galaxy</i><br><i>Description=[Dwarf irregular galaxies]</i>                                                                                                                                                                                                                                                                                                          | Epoch of Position: 2000                                                                                                           |               |
|               | (4) | UGC-8091     | RA: 12 58 39.3548 (194.6639783d)<br>Dec: +14 13 8.05 (14.21890d)<br>Equinox: J2000<br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i><br><i>Category=Galaxy</i><br><i>Description=[Dwarf irregular galaxies]</i> | Proper Motion RA: -1.69 mas/yr<br>Proper Motion Dec: -0.8420000312980847 mas/yr<br>Parallax: 2.351E-4"<br>Epoch of Position: 2000 |               |
|               | (5) | LEDA-2807150 | RA: 13 54 33.3560 (208.6389833d)<br>Dec: +04 14 39.66 (4.24435d)<br>Equinox: J2000<br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=Galaxy</i><br><i>Description=[Dwarf galaxies]</i>                                                                                                                                                                                                                                                                                                                     | Epoch of Position: 2000                                                                                                           |               |
|               | (6) | IC-5152      | RA: 22 02 44.4858 (330.6853575d)<br>Dec: -51 18 11.63 (-51.30323d)<br>Equinox: J2000<br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=Galaxy</i><br><i>Description=[Dwarf galaxies]</i>                                                                                                                                                                                                                                                                                                                   | Epoch of Position: 2000                                                                                                           |               |
|               | (7) | NGC-6789     | RA: 19 16 40.4223 (289.1684263d)<br>Dec: +63 58 30.92 (63.97526d)<br>Equinox: J2000<br><i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i><br><i>Category=Galaxy</i><br><i>Description=[Compact dwarf galaxy]</i>                                                                                                                                                                                                                                                                                                              | Epoch of Position: 2000                                                                                                           |               |

# Proposal 7232 - Observation 1 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

|                      |                                                                                                                                                                |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|-----------------|------------|--------------------------|----------------------|---------------|---------------------|--------------------|------------------------------|
| Observation          | Proposal 7232, Observation 1: DDO190                                                                                                                           |                     |                                                                                     |                 |            |                          |                      |               |                     |                    | Tue Mar 11 20:08:27 GMT 2025 |
|                      | Diagnostic Status: Warning                                                                                                                                     |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Observing Template: NIRCcam Imaging                                                                                                                            |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Comments: ORIENT limits are used to align the NIRCAM FOV with the galaxy major axis.  Schedulability is minimally impacted (>37 day visibility window).        |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Diagnostics          | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                    |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Fixed Targets        | #                                                                                                                                                              | Name                | Target Coordinates                                                                  |                 |            | Targ. Coord. Corrections |                      |               | Miscellaneous       |                    |                              |
|                      | (1)                                                                                                                                                            | UGC-9240            | RA: 14 24 43.2100 (216.1800417d)<br>Dec: +44 31 41.80 (44.52828d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |                      |               |                     |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.<br>Category=Galaxy<br>Description=[Dwarf irregular galaxies] |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      |                                                                                                                                                                |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Template             | Module                                                                                                                                                         |                     | Subarray                                                                            |                 |            |                          | Target Placement     |               |                     |                    |                              |
|                      | ALL                                                                                                                                                            |                     | FULL                                                                                |                 |            |                          | Module B (B4 corner) |               |                     |                    |                              |
| Dithers              | #                                                                                                                                                              | Primary Dither Type |                                                                                     | Primary Dithers |            | Subpixel Dither Type     |                      | Dither Size   |                     | Subpixel Positions |                              |
|                      | 1                                                                                                                                                              | NONE                |                                                                                     |                 |            | STANDARD                 |                      |               |                     | 5                  |                              |
| Spectral Elements    | #                                                                                                                                                              | Short Filter        | Long Filter                                                                         | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations   | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID   |                              |
|                      | 1                                                                                                                                                              | F090W               | F360M                                                                               | MEDIUM8         | 5          | 3                        | 15                   | 5             | 7837.842            |                    |                              |
|                      | 2                                                                                                                                                              | F150W               | F480M                                                                               | MEDIUM8         | 5          | 2                        | 10                   | 5             | 5207.333            |                    |                              |
| Special Requirements | Aperture PA Range 180.05262691 to 215.05262691 Degrees (V3 180.0 to 215.0)<br>Fiducial Point Override NRCBS_FULL                                               |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |

# Proposal 7232 - Observation 2 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

|                      |                                                                                                                                                                              |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|-----------------|------------|--------------------------|----------------------|---------------|---------------------|--------------------|------------------------------|
| Observation          | Proposal 7232, Observation 2: IC4662                                                                                                                                         |                     |                                                                                     |                 |            |                          |                      |               |                     |                    | Tue Mar 11 20:08:27 GMT 2025 |
|                      | Diagnostic Status: Warning                                                                                                                                                   |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Observing Template: NIRCcam Imaging                                                                                                                                          |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Comments: POS TARG offset and ORIENT limits set to avoid bright star 1.2' NW of target and bright stars 3.4' NNE of target.                                                  |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Diagnostics          | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                  |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Fixed Targets        | #                                                                                                                                                                            | Name                | Target Coordinates                                                                  |                 |            | Targ. Coord. Corrections |                      |               | Miscellaneous       |                    |                              |
|                      | (2)                                                                                                                                                                          | IC-4662             | RA: 17 47 9.1920 (266.7883000d)<br>Dec: -64 38 47.90 (-64.64664d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |                      |               |                     |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                            |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Category=Galaxy<br>Description=[Compact dwarf galaxy]                                                                                                                        |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Template             | Module                                                                                                                                                                       |                     | Subarray                                                                            |                 |            |                          | Target Placement     |               |                     |                    |                              |
|                      | ALL                                                                                                                                                                          |                     | FULL                                                                                |                 |            |                          | Module B (B4 corner) |               |                     |                    |                              |
| Dithers              | #                                                                                                                                                                            | Primary Dither Type |                                                                                     | Primary Dithers |            | Subpixel Dither Type     |                      | Dither Size   |                     | Subpixel Positions |                              |
|                      | 1                                                                                                                                                                            | NONE                |                                                                                     |                 |            | STANDARD                 |                      |               |                     | 4                  |                              |
| Spectral Elements    | #                                                                                                                                                                            | Short Filter        | Long Filter                                                                         | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations   | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID   |                              |
|                      | 1                                                                                                                                                                            | F090W               | F360M                                                                               | MEDIUM8         | 5          | 3                        | 12                   | 4             | 6270.274            |                    |                              |
|                      | 2                                                                                                                                                                            | F150W               | F480M                                                                               | MEDIUM8         | 5          | 2                        | 8                    | 4             | 4165.867            |                    |                              |
| Special Requirements | Aperture PA Range 310.05262691 to 255.05262691 Degrees (V3 310.0 to 255.0)<br>Offset 22.74342860101085 arcsec, 2.57462458480637 arcsec<br>Fiducial Point Override NRCBS_FULL |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |

# Proposal 7232 - Observation 3 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

|                      |                                                                                                                                                                                                |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|-----------------|------------|--------------------------|----------------------|---------------|---------------------|--------------------|------------------------------|
| Observation          | Proposal 7232, Observation 3: DDO187                                                                                                                                                           |                     |                                                                                     |                 |            |                          |                      |               |                     |                    | Tue Mar 11 20:08:27 GMT 2025 |
|                      | Diagnostic Status: Warning                                                                                                                                                                     |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Observing Template: NIRCcam Imaging                                                                                                                                                            |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Comments: ORIENT constraints set to align NIRCAM FOV with galaxy major axis while avoiding bright star 3.3' to SSW.                                                                            |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Diagnostics          | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                    |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Fixed Targets        | #                                                                                                                                                                                              | Name                | Target Coordinates                                                                  |                 |            | Targ. Coord. Corrections |                      |               | Miscellaneous       |                    |                              |
|                      | (3)                                                                                                                                                                                            | UGC-9128            | RA: 14 15 57.0834 (213.9878475d)<br>Dec: +23 03 29.87 (23.05830d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |                      |               |                     |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                              |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
|                      | Category=Galaxy<br>Description=[Dwarf irregular galaxies]                                                                                                                                      |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |
| Template             | Module                                                                                                                                                                                         |                     | Subarray                                                                            |                 |            |                          | Target Placement     |               |                     |                    |                              |
|                      | ALL                                                                                                                                                                                            |                     | FULL                                                                                |                 |            |                          | Module B (B4 corner) |               |                     |                    |                              |
| Dithers              | #                                                                                                                                                                                              | Primary Dither Type |                                                                                     | Primary Dithers |            | Subpixel Dither Type     |                      | Dither Size   |                     | Subpixel Positions |                              |
|                      | 1                                                                                                                                                                                              | NONE                |                                                                                     |                 |            | STANDARD                 |                      |               |                     | 4                  |                              |
| Spectral Elements    | #                                                                                                                                                                                              | Short Filter        | Long Filter                                                                         | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations   | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID   |                              |
|                      | 1                                                                                                                                                                                              | F090W               | F360M                                                                               | MEDIUM8         | 5          | 2                        | 8                    | 4             | 4165.867            |                    |                              |
|                      | 2                                                                                                                                                                                              | F150W               | F480M                                                                               | MEDIUM8         | 6          | 1                        | 4                    | 4             | 2490.931            |                    |                              |
| Special Requirements | Aperture PA Range 130.05262691 to 145.05262691 Degrees (V3 130.0 to 145.0)<br>Aperture PA Range 300.05262691 to 320.05262691 Degrees (V3 300.0 to 320.0)<br>Fiducial Point Override NRCBS_FULL |                     |                                                                                     |                 |            |                          |                      |               |                     |                    |                              |

# Proposal 7232 - Observation 4 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

|                      |                                                                                                                                                                                                                                                                                                                                                              |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------|-----------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------|---------------------|--------------------|------------------------------|
| Observation          | Proposal 7232, Observation 4: GR8                                                                                                                                                                                                                                                                                                                            |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    | Tue Mar 11 20:08:27 GMT 2025 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                                                                                                                                                                   |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |
|                      | Observing Template: NIRCам Imaging                                                                                                                                                                                                                                                                                                                           |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |
|                      | Comments: Orient constraints used to align NIRCAM FOV with galaxy major axis. JWST ETC reveals background variability of 6.1%, but we choose not to impose a background limited constraint to maintain schedulability.                                                                                                                                       |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |
| Diagnostics          | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                                                                                  |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |
| Fixed Targets        | #                                                                                                                                                                                                                                                                                                                                                            | Name                | Target Coordinates                                                                 |                 |            | Targ. Coord. Corrections                                                                                                          |                      |               | Miscellaneous       |                    |                              |
|                      | (4)                                                                                                                                                                                                                                                                                                                                                          | UGC-8091            | RA: 12 58 39.3548 (194.6639783d)<br>Dec: +14 13 8.05 (14.21890d)<br>Equinox: J2000 |                 |            | Proper Motion RA: -1.69 mas/yr<br>Proper Motion Dec: -0.8420000312980847 mas/yr<br>Parallax: 2.351E-4"<br>Epoch of Position: 2000 |                      |               |                     |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                                                                                                                                                                                            |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |
|                      | SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.<br>Category=Galaxy<br>Description=[Dwarf irregular galaxies] |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |
| Template             | Module                                                                                                                                                                                                                                                                                                                                                       |                     | Subarray                                                                           |                 |            |                                                                                                                                   | Target Placement     |               |                     |                    |                              |
|                      | ALL                                                                                                                                                                                                                                                                                                                                                          |                     | FULL                                                                               |                 |            |                                                                                                                                   | Module B (B4 corner) |               |                     |                    |                              |
| Dithers              | #                                                                                                                                                                                                                                                                                                                                                            | Primary Dither Type |                                                                                    | Primary Dithers |            | Subpixel Dither Type                                                                                                              |                      | Dither Size   |                     | Subpixel Positions |                              |
|                      | 1                                                                                                                                                                                                                                                                                                                                                            | NONE                |                                                                                    |                 |            | STANDARD                                                                                                                          |                      |               |                     | 4                  |                              |
| Spectral Elements    | #                                                                                                                                                                                                                                                                                                                                                            | Short Filter        | Long Filter                                                                        | Readout Pattern | Groups/Int | Integrations/Exp                                                                                                                  | Total Integrations   | Total Dithers | Total Exposure Time | ETC Wkbk.Calc ID   |                              |
|                      | 1                                                                                                                                                                                                                                                                                                                                                            | F090W               | F360M                                                                              | MEDIUM8         | 5          | 2                                                                                                                                 | 8                    | 4             | 4165.867            |                    |                              |
|                      | 2                                                                                                                                                                                                                                                                                                                                                            | F150W               | F480M                                                                              | MEDIUM8         | 6          | 1                                                                                                                                 | 4                    | 4             | 2490.931            |                    |                              |
| Special Requirements | Aperture PA Range 115.05262691 to 140.05262691 Degrees (V3 115.0 to 140.0)<br>Aperture PA Range 295.05262691 to 310.05262691 Degrees (V3 295.0 to 310.0)<br>Fiducial Point Override NRCBS_FULL                                                                                                                                                               |                     |                                                                                    |                 |            |                                                                                                                                   |                      |               |                     |                    |                              |

# Proposal 7232 - Observation 5 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

|                      |                                                                                                                                                                                                                                                                                                                                                                                               |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------|-----------------|------------|--------------------------|--------------------|----------------------|---------------------|--------------------|------------------------------|
| Observation          | Proposal 7232, Observation 5: KKH86                                                                                                                                                                                                                                                                                                                                                           |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    | Tue Mar 11 20:08:27 GMT 2025 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                                                                                                                                                                                                    |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |
|                      | Observing Template: NIRCcam Imaging                                                                                                                                                                                                                                                                                                                                                           |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |
|                      | Comments: For some ORIENTs, bright foreground stars would fall on one of the NIRCcam modules, but implementing ORIENT constraints results in small visibility windows and conflicts with the micrometeoroid zone. A combination of POS TARG offsets and/or ORIENT limitations may be used to avoid these bright sources once an observation window is confirmed. Background variability 9.3%. |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |
| Diagnostics          | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                                                                                                                   |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |
| Fixed Targets        | #                                                                                                                                                                                                                                                                                                                                                                                             | Name                | Target Coordinates                                                                 |                 |            | Targ. Coord. Corrections |                    |                      | Miscellaneous       |                    |                              |
|                      | (5)                                                                                                                                                                                                                                                                                                                                                                                           | LEDA-2807150        | RA: 13 54 33.3560 (208.6389833d)<br>Dec: +04 14 39.66 (4.24435d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |                    |                      |                     |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                                                                                                                                                                                                                             |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |
|                      | Category=Galaxy<br>Description=[Dwarf galaxies]                                                                                                                                                                                                                                                                                                                                               |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |
| Template             | Module                                                                                                                                                                                                                                                                                                                                                                                        |                     |                                                                                    | Subarray        |            |                          |                    | Target Placement     |                     |                    |                              |
|                      | ALL                                                                                                                                                                                                                                                                                                                                                                                           |                     |                                                                                    | FULL            |            |                          |                    | Module B (B4 corner) |                     |                    |                              |
| Dithers              | #                                                                                                                                                                                                                                                                                                                                                                                             | Primary Dither Type |                                                                                    | Primary Dithers |            | Subpixel Dither Type     |                    | Dither Size          |                     | Subpixel Positions |                              |
|                      | 1                                                                                                                                                                                                                                                                                                                                                                                             | NONE                |                                                                                    |                 |            | STANDARD                 |                    |                      |                     | 4                  |                              |
| Spectral Elements    | #                                                                                                                                                                                                                                                                                                                                                                                             | Short Filter        | Long Filter                                                                        | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations | Total Dithers        | Total Exposure Time | ETC Wkbk.Calc ID   |                              |
|                      | 1                                                                                                                                                                                                                                                                                                                                                                                             | F090W               | F360M                                                                              | MEDIUM8         | 6          | 3                        | 12                 | 4                    | 7558.686            |                    |                              |
|                      | 2                                                                                                                                                                                                                                                                                                                                                                                             | F150W               | F480M                                                                              | MEDIUM8         | 7          | 2                        | 8                  | 4                    | 5883.75             |                    |                              |
| Special Requirements | Background Limited. Background no more than 10th percentile above minimum<br>Fiducial Point Override NRCBS_FULL                                                                                                                                                                                                                                                                               |                     |                                                                                    |                 |            |                          |                    |                      |                     |                    |                              |



# Proposal 7232 - Observation 6 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

|                      |                                                                                                                                                                                                                                                              |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------|-----------------|------------|--------------------------|--------------------|---------------|----------------------|--------------------|------------------------------|
| Observation          | Proposal 7232, Observation 6: IC5152                                                                                                                                                                                                                         |                     |                                                                                      |                 |            |                          |                    |               |                      |                    | Tue Mar 11 20:08:27 GMT 2025 |
|                      | Diagnostic Status: Warning                                                                                                                                                                                                                                   |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |
|                      | Observing Template: NIRCAM Imaging                                                                                                                                                                                                                           |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |
|                      | Comments: V3PA constraints and POS TARG offset used to align NIRCAM FOV with galaxy major axis and avoid bright star 1' NW of target.                                                                                                                        |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |
| Diagnostics          | (Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                  |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |
| Fixed Targets        | #                                                                                                                                                                                                                                                            | Name                | Target Coordinates                                                                   |                 |            | Targ. Coord. Corrections |                    |               | Miscellaneous        |                    |                              |
|                      | (6)                                                                                                                                                                                                                                                          | IC-5152             | RA: 22 02 44.4858 (330.6853575d)<br>Dec: -51 18 11.63 (-51.30323d)<br>Equinox: J2000 |                 |            | Epoch of Position: 2000  |                    |               |                      |                    |                              |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.                                                                                                                                                            |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |
|                      | Category=Galaxy<br>Description=[Dwarf galaxies]                                                                                                                                                                                                              |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |
| Template             | Module                                                                                                                                                                                                                                                       |                     |                                                                                      |                 | Subarray   |                          |                    |               | Target Placement     |                    |                              |
|                      | ALL                                                                                                                                                                                                                                                          |                     |                                                                                      |                 | FULL       |                          |                    |               | Module B (B4 corner) |                    |                              |
| Dithers              | #                                                                                                                                                                                                                                                            | Primary Dither Type |                                                                                      | Primary Dithers |            | Subpixel Dither Type     |                    | Dither Size   |                      | Subpixel Positions |                              |
|                      | 1                                                                                                                                                                                                                                                            | NONE                |                                                                                      |                 |            | STANDARD                 |                    |               |                      | 4                  |                              |
| Spectral Elements    | #                                                                                                                                                                                                                                                            | Short Filter        | Long Filter                                                                          | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations | Total Dithers | Total Exposure Time  | ETC Wkbk.Calc ID   |                              |
|                      | 1                                                                                                                                                                                                                                                            | F090W               | F360M                                                                                | MEDIUM8         | 6          | 1                        | 4                  | 4             | 2490.931             |                    |                              |
|                      | 2                                                                                                                                                                                                                                                            | F150W               | F480M                                                                                | MEDIUM8         | 5          | 1                        | 4                  | 4             | 2061.46              |                    |                              |
| Special Requirements | Aperture PA Range 170.05262691 to 185.05262691 Degrees (V3 170.0 to 185.0)<br>Aperture PA Range 345.05262691 to 15.05262691 Degrees (V3 345.0 to 15.0)<br>Offset -10.544182045002394 arcsec, 12.627802337175018 arcsec<br>Fiducial Point Override NRCBS_FULL |                     |                                                                                      |                 |            |                          |                    |               |                      |                    |                              |

# Proposal 7232 - Observation 7 - Orphan Dwarfs: Benchmarking Low-Mass Galaxy Evolution in an Isolated Environment

|                      |                                                                                                   |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
|----------------------|---------------------------------------------------------------------------------------------------|---------------------|--|----------------------------------|-----------------|------------|--------------------------|--------------------|----------------------|---------------------|------------------------------|--|
| Observation          | Proposal 7232, Observation 7: NGC6789                                                             |                     |  |                                  |                 |            |                          |                    |                      |                     | Tue Mar 11 20:08:28 GMT 2025 |  |
|                      | Diagnostic Status: Warning                                                                        |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
|                      | Observing Template: NIRCcam Imaging                                                               |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
|                      | Comments: Background variability 5.8%                                                             |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
| Diagnostics          | (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.       |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
| Fixed Targets        | #                                                                                                 | Name                |  | Target Coordinates               |                 |            | Targ. Coord. Corrections |                    |                      | Miscellaneous       |                              |  |
|                      | (7)                                                                                               | NGC-6789            |  | RA: 19 16 40.4223 (289.1684263d) |                 |            | Epoch of Position: 2000  |                    |                      |                     |                              |  |
|                      |                                                                                                   |                     |  | Dec: +63 58 30.92 (63.97526d)    |                 |            |                          |                    |                      |                     |                              |  |
|                      |                                                                                                   |                     |  | Equinox: J2000                   |                 |            |                          |                    |                      |                     |                              |  |
|                      | Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
|                      | Category=Galaxy                                                                                   |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
|                      | Description=[Compact dwarf galaxy]                                                                |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
| Template             | Module                                                                                            |                     |  |                                  | Subarray        |            |                          |                    | Target Placement     |                     |                              |  |
|                      | ALL                                                                                               |                     |  |                                  | FULL            |            |                          |                    | Module B (B4 corner) |                     |                              |  |
| Dithers              | #                                                                                                 | Primary Dither Type |  |                                  | Primary Dithers |            | Subpixel Dither Type     |                    | Dither Size          |                     | Subpixel Positions           |  |
|                      | 1                                                                                                 | NONE                |  |                                  |                 |            | STANDARD                 |                    |                      |                     | 4                            |  |
| Spectral Elements    | #                                                                                                 | Short Filter        |  | Long Filter                      | Readout Pattern | Groups/Int | Integrations/Exp         | Total Integrations | Total Dithers        | Total Exposure Time | ETC Wkbk.Calc ID             |  |
|                      | 1                                                                                                 | F090W               |  | F360M                            | MEDIUM8         | 8          | 4                        | 16                 | 4                    | 13528.33            |                              |  |
|                      | 2                                                                                                 | F150W               |  | F480M                            | MEDIUM8         | 8          | 3                        | 12                 | 4                    | 10135.511           |                              |  |
| Special Requirements | Background Limited. Background no more than 10th percentile above minimum                         |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |
|                      | Fiducial Point Override NRCBS_FULL                                                                |                     |  |                                  |                 |            |                          |                    |                      |                     |                              |  |